Test Cases Document

Contents

Test Cases Document 1

1. Test Case Document – Environment Class 1

2. Test Case Document – Robot Class 1

3. Test Case Document – Performance Class 3

4. Test Case Document – Integration Class 3

1. Test Case Document – Environment Class

Test Case ID	Test Case Objective	Steps	Input Data	Expected Result	Actual Result	Status
TC1	Initialize Environment	1. Implement the Environment class.2. Create an instance: env = Environment(size=100, bio_hazard_number=5000).3. Verify grid, non_accessible, and bio_hazard lists exist.	Environment(size=100, bio_hazard_number=5000)	Grid of 100×100 is created. Lists non_accessible and bio_hazard exist.	As Expected	Pass
TC2	Generate Non-Accessible Areas	1. Initialize Environment.2. Check non_accessible list.3. Confirm all marked cells are -1.	Default non_accessible_info	All non-accessible coordinates are marked -1 in the grid.	As Expected	Pass
TC3	Generate Bio-Hazards	1. Initialize Environment.2. Call bio-hazard generation with bio_hazard_number=5.3. Verify bio_hazard list length and grid updates.	bio_hazard_number=5	bio_hazard list length = 5; each bio-hazard coordinate in grid = 1 and accessible.	As Expected	Pass
TC4	Check Accessibility	1. Initialize Environment.2. Call is_accessible(x, y) for selected coordinates.3. Verify return value.	Coordinates (x, y)	Returns True if accessible, False if non-accessible or out-of-bounds.	As Expected	Pass
TC5	Check Bio-Hazard	1. Initialize Environment.2. Call has_bio_hazard(x, y).3. Verify return value.	Coordinates (x, y)	Returns True if bio-hazard present, False otherwise.	As Expected	Pass
TC6	Boundaries of Non-Accessible Area	1. Initialize Environment.2. Iterate over non_accessible coordinates.3. Verify all 0 <= x,y < size.	Non-accessible areas	All coordinates are within grid boundaries.	As Expected	Pass
TC7	Show Environment Output	1. Initialize Environment.2. Call show_environment().3. Verify printed output lists coordinates.	Environment initialized	Lists accessible, non-accessible, and bio-hazard coordinates printed.	As Expected	Pass
TC8	Bio-Hazard Avoids Non-Accessible	1. Initialize Environment.2. Generate bio-hazards.3. Check that none overlap with non-accessible cells.	Default environment	No bio-hazard coordinate overlaps with non-accessible coordinates.	As Expected	Pass

2. Test Case Document – Robot Class

Test Case ID	Test Case Objective	Steps	Input Data	Expected Result	Actual Result	Status
TC1	Initialize	1. Ensure Environment is implemented.2.	Robot(env)	Robot placed on	As	Pass

	Robot	Create robot = Robot(env).3. Verify robot is placed on accessible cell; visited list initialized.		accessible cell; visited contains starting position; counters initialized.	Expecte d	
TC2	Perceive on empty cell	1. Initialize Robot.2. Move to a cell without bio-hazard.3. Call perceive().4. Check biohazard_checks increment; no new collections; cleaning_actions unchanged.	Empty cell	biohazard_checks incremented; cleaning_actions unchanged.	As Expecte d	Pass
TC3	Perceive on bio-hazard cell	1. Initialize Robot.2. Move to cell with bio-hazard.3. Call perceive().4. Verify collection and cleaning_actions increment.	Bio-hazard cell	biohazard_checks incremented; cell collected; cleaning_actions incremented; grid updated to 0.	As Expecte d	Pass
TC4	Collect bio-hazard	1. Robot placed on bio-hazard.2. Call collect().3. Verify object added to collected_objects and cell cleared.	Bio-hazard cell	Object added to collected_objects; cell cleared; cleaning_actions incremented.	As Expecte d	Pass
TC5	Can move to accessible unvisited cell	1. Initialize Robot.2. Call can_move(x, y) on accessible, unvisited cell.3. Check return value and increment of accessibility_checks.	Accessible & unvisited	Returns True; accessibility_check s incremented.	As Expecte d	Pass
TC6	Can move to accessible visited cell	1. Initialize Robot.2. Move to previously visited cell.3. Call can_move(x, y, allow_visited=False) and allow_visited=True.	Visited cell	Returns False if allow_visited=Fals e; True if allow_visited=True .	As Expecte d	Pass
TC7	Can move to non-accessible cell	1. Initialize Robot.2. Call can_move(x, y) on non-accessible cell.3. Verify it returns False.	Non-accessible cell	Returns False.	As Expecte d	Pass
TC8	Can move to out-of-bounds cell	1. Initialize Robot.2. Call can_move(x, y) with x or y out of grid.3. Verify it returns False.	Out-of-bounds cell	Returns False.	As Expecte d	Pass
TC9	Move in valid direction	1. Initialize Robot.2. Call move("UP"/"DOWN"/"LEFT"/"RIGHT").3. Verify position updated, visited updated, movement_actions incremented, perception triggered.	Direction	Robot moves and updates state.	As Expecte d	Pass
TC1 0	Move in invalid direction	1. Initialize Robot.2. Call move("INVALID").3. Verify it returns False and state unchanged.	Direction	Returns False; position and move_history unchanged.	As Expecte d	Pass
TC1 1	Move into non-accessible cell	1. Initialize Robot.2. Call move() into non-accessible cell.3. Verify move fails, position unchanged.	Non-accessible direction	Move fails; position unchanged.	As Expecte d	Pass
TC1 2	Random move returns boolean	1. Initialize Robot.2. Call random_move().3. Verify return type True/False if moved; None if stuck.	Random move	Returns True/False or None.	As Expecte d	Pass
TC1 3	Random move updates state	1. Initialize Robot.2. Call random_move().3. Verify move_history, visited, and movement_actions updated.	Random move	State updated correctly.	As Expecte d	Pass
TC1 4	Random move when stuck	1. Block Robot.2. Call random_move().3. Verify returns None, no state change.	Robot blocked	Returns None; state unchanged.	As Expecte d	Pass
TC1 5	Robot collects on move	1. Move Robot onto bio-hazard cell.2. Verify object collected, cleaning_actions incremented, grid updated.	Bio-hazard	Object collected; grid updated; cleaning_actions incremented.	As Expecte d	Pass
TC1	Get	1. Initialize Robot.2. Call	Robot initialized	Returns (x, y).	As	Pass

6	current position	get_position().3. Verify returned coordinates.			Expected	
TC17	Get move history	1. Robot performs moves.2. Call get_move_history().3. Verify returned sequence matches moves.	Robot moved	Returns correct move sequence.	As Expected	Pass
TC18	Get collected objects	1. Robot collects objects.2. Call get_collected_objects().3. Verify list matches collected objects.	Robot collected objects	Returns collected objects list.	As Expected	Pass
TC19	Simulation summary prints stats	1. Robot performs actions.2. Call simulation_summary().3. Verify printed metrics.	Robot with actions	Prints accessibility_checks, biohazard_checks, total percepts, movement_actions, cleaning_actions.	As Expected	Pass
TC20	Multiple random moves avoid revisits	1. Robot performs 5 random_move() calls.2. Verify number of visited cells and no revisits counted improperly.	Random moves	Number of visited >= moves+1; no revisits counted improperly.	As Expected	Pass
TC21	Can move after collection	1. Robot collects bio-hazard.2. Perform random_move().3. Verify move succeeds or stuck but no errors.	After collection	Robot moves successfully or is stuck; no errors.	As Expected	Pass
TC22	Move history accuracy	1. Robot moves multiple directions.2. Call get_move_history().3. Verify sequence matches actual moves.	Move sequence ["UP","RIGHT","DOWN","LEFT"]	Move history matches successful moves.	As Expected	Pass
TC23	Robot always placed on accessible cell	1. Initialize Robot 10 times in same Environment.2. Verify starting position is always accessible.	10 initializations	Initial position never non-accessible.	As Expected	Pass

3. Test Case Document – Performance Class

Test Case ID	Test Case Objective	Steps	Input Data	Expected Result	Actual Result	Status
TC1	Performance Initialization	1. Create Robot.2. Initialize Performance with robot: perf = Performance(robot).3. Verify reference to robot.	Robot created	Performance object references correct robot.	As Expected	Pass
TC2	Total Collected Method	1. Robot collects objects.2. Call perf.total_collected().3. Verify count matches robot.collected_objects.	Robot collected objects	Returns correct number of collected objects.	As Expected	Pass
TC3	Total Moves Method	1. Robot performs moves.2. Call perf.total_moves().3. Verify count matches robot.move_history.	Robot moves	Returns correct number of moves.	As Expected	Pass
TC4	Efficiency Calculation	1. Robot has collections and moves.2. Call perf.efficiency().3. Verify efficiency = collections / moves.	Robot state	Correct efficiency calculation.	As Expected	Pass
TC5	Efficiency With Zero Moves	1. Robot has no moves but has collections.2. Call perf.efficiency().3. Verify return value.	Empty move_history	Returns 0.	As Expected	Pass
TC6	Summary Method	1. Call perf.summary().2. Verify all metrics printed correctly.	Performance object	Prints collected objects, total moves, total actions, efficiency.	As Expected	Pass

4. Test Case Document – Integration Class

Test Case ID	Test Case Objective	Steps	Input Data	Expected Result	Actual Result	Status
TC1	Full simulation: robot moves, collects, performance tracked	Robot and Performance objects created	move_limit = 50	Robot moves recorded; collected objects counted; efficiency 0–1	Robot moved, objects collected, efficiency calculated correctly	Pass

TC2	Robot never moves into non-accessible cells	Robot created in environment	Random moves	Robot only moves to accessible cells	Robot stayed within accessible cells	Pass
TC3	Collected objects removed from environment grid	Robot created; bio-hazard placed at robot's position	Robot position = (x, y) with bio-hazard	Robot collects object; grid updated to 0	Object collected; grid cleared	Pass

```
PS D:\3-2\Artificial Intelligence\Lab> python -m pytest "Lab 2/" -v
platform win32 -- Python 3.12.3, pytest-9.0.2, pluggy-1.6.0 -- C:\Program Files\Python312\python.exe
cachedir: .pytest_cache
rootdir: D:\3-2\Artificial Intelligence\Lab
plugins: anyio-4.4.0
collected 51 items

Lab 2/test_action_perception.py::TestRobot::test_robot_initialization PASSED                                  [  1%]
Lab 2/test_action_perception.py::TestRobot::test_perceive_no_biohazard PASSED                                 [  3%]
Lab 2/test_action_perception.py::TestRobot::test_perceive_with_biohazard PASSED                               [  5%]
Lab 2/test_action_perception.py::TestRobot::test_collect PASSED                                               [  7%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_accessible_unvisited PASSED                         [  9%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_accessible_visited PASSED                           [ 11%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_accessible_visited_allowed PASSED                   [ 13%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_non_accessible PASSED                               [ 15%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[-1-0] PASSED                          [ 17%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[0--1] PASSED                          [ 19%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[10-0] PASSED                          [ 21%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[0-10] PASSED                          [ 23%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[-5--5] PASSED                         [ 25%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[15-15] PASSED                         [ 27%]
Lab 2/test_action_perception.py::TestRobot::test_move_valid_direction PASSED                                  [ 29%]
Lab 2/test_action_perception.py::TestRobot::test_move_invalid_direction PASSED                                [ 31%]
Lab 2/test_action_perception.py::TestRobot::test_move_into_non_accessible PASSED                              [ 33%]
Lab 2/test_action_perception.py::TestRobot::test_random_move_returns_boolean PASSED                           [ 35%]
Lab 2/test_action_perception.py::TestRobot::test_random_move_updates_state PASSED                             [ 37%]
Lab 2/test_action_perception.py::TestRobot::test_random_move_stuck PASSED                                     [ 39%]
Lab 2/test_action_perception.py::TestRobot::test_robot_collects_on_move PASSED                                [ 41%]
Lab 2/test_action_perception.py::TestRobot::test_get_position PASSED                                          [ 43%]
Lab 2/test_action_perception.py::TestRobot::test_get_move_history PASSED                                      [ 45%]
Lab 2/test_action_perception.py::TestRobot::test_get_collected_objects PASSED                                 [ 47%]
Lab 2/test_action_perception.py::TestRobot::test_simulation_summary PASSED                                    [ 49%]
Lab 2/test_action_perception.py::TestRobot::test_multiple_moves_no_revisits PASSED                            [ 50%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_after_collection PASSED                             [ 52%]
Lab 2/test_action_perception.py::TestRobot::test_move_history_accuracy PASSED                                 [ 54%]
Lab 2/test_action_perception.py::TestRobot::test_robot_always_accessible PASSED                               [ 56%]
```

```
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[-1-0] PASSED                          [ 17%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[0--1] PASSED                          [ 19%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[10-0] PASSED                          [ 21%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[0-10] PASSED                          [ 23%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[-5--5] PASSED                         [ 25%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_out_of_bounds[15-15] PASSED                         [ 27%]
Lab 2/test_action_perception.py::TestRobot::test_move_valid_direction PASSED                                  [ 29%]
Lab 2/test_action_perception.py::TestRobot::test_move_invalid_direction PASSED                                [ 31%]
Lab 2/test_action_perception.py::TestRobot::test_move_into_non_accessible PASSED                              [ 33%]
Lab 2/test_action_perception.py::TestRobot::test_random_move_returns_boolean PASSED                           [ 35%]
Lab 2/test_action_perception.py::TestRobot::test_random_move_updates_state PASSED                             [ 37%]
Lab 2/test_action_perception.py::TestRobot::test_random_move_stuck PASSED                                     [ 39%]
Lab 2/test_action_perception.py::TestRobot::test_robot_collects_on_move PASSED                                [ 41%]
Lab 2/test_action_perception.py::TestRobot::test_get_position PASSED                                          [ 43%]
Lab 2/test_action_perception.py::TestRobot::test_get_move_history PASSED                                      [ 45%]
Lab 2/test_action_perception.py::TestRobot::test_get_collected_objects PASSED                                 [ 47%]
Lab 2/test_action_perception.py::TestRobot::test_simulation_summary PASSED                                    [ 49%]
Lab 2/test_action_perception.py::TestRobot::test_multiple_moves_no_revisits PASSED                            [ 50%]
Lab 2/test_action_perception.py::TestRobot::test_can_move_after_collection PASSED                             [ 52%]
Lab 2/test_action_perception.py::TestRobot::test_move_history_accuracy PASSED                                 [ 54%]
Lab 2/test_action_perception.py::TestRobot::test_robot_always_accessible PASSED                               [ 56%]
                    [ 80%]
Lab 2/test_integration.py::TestIntegration::test_full_simulation PASSED                                       [ 82%]
Lab 2/test_integration.py::TestIntegration::test_robot_respects_boundaries PASSED                             [ 84%]
Lab 2/test_integration.py::TestIntegration::test_collected_objects_cleared PASSED                             [ 86%]
Lab 2/test_integration.py::TestIntegration::test_performance_summary PASSED                                   [ 88%]
Lab 2/test_performance.py::TestPerformance::test_performance_initialization PASSED                            [ 90%]
Lab 2/test_performance.py::TestPerformance::test_total_collected PASSED                                       [ 92%]
Lab 2/test_performance.py::TestPerformance::test_total_moves PASSED                                           [ 94%]
Lab 2/test_performance.py::TestPerformance::test_efficiency_calculation PASSED                                [ 96%]
Lab 2/test_performance.py::TestPerformance::test_efficiency_zero_moves PASSED                                 [ 98%]
Lab 2/test_performance.py::TestPerformance::test_summary_output PASSED                                        [100%]

============================================================== 51 passed in 0.41s ==============================================================
PS D:\3-2\Artificial Intelligence\Lab>
```